playground

2025-12-02

```
library(knitr)
```

```
## Warning: package 'knitr' was built under R version 4.5.1
```

```
library(haven)
```

```
## Warning: package 'haven' was built under R version 4.5.1
```

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(car)
```

```
## Warning: package 'car' was built under R version 4.5.1
```

```
## Loading required package: carData
```

```
## Warning: package 'carData' was built under R version 4.5.1
```

```
##
## Attaching package: 'car'
```

```
## The following object is masked from 'package:dplyr':
##
##     recode
```

```
library(survey)
```

```
## Warning: package 'survey' was built under R version 4.5.2
```

```
## Loading required package: grid
```

```
## Loading required package: Matrix
```

```
## Loading required package: survival
```

```
##
## Attaching package: 'survey'
```

```
## The following object is masked from 'package:graphics':
##
##     dotchart
```

```r
library(srvyr)
```

```
## Warning: package 'srvyr' was built under R version 4.5.2
```

```
##
## Attaching package: 'srvyr'
```

```
## The following object is masked from 'package:stats':
##
##     filter
```

Read datasets

```
#Nacist data a z nekterych vybrat jen nektere promenne

#Demographics, RIDAGEYR - vek, RIDEXPRG - pregnant, DMDEDUC2 - education level, RIAGENDR - gender, INDFMPIR - ratio of family income to poverty
#RIDSTATR - both interview and MEC examination (2 - both, 1 - only interview)
demo_data = read_xpt("./data/DEMO_H.xpt") |> select(SEQN,RIDAGEYR,RIDEXPRG,DMDEDUC2,RIAGENDR,INDFMPIR,WTMEC2YR, SDMVPSU,SDMVSTRA)
#Reproductive health data, RHQ031 - had regular periods in past 12 months, RHQ074 - tried for a yer to become pregnant, RHQ076 - seen DR unable to become pregnant, RHQ131 - ever been pregnant,
#RHQ160 - how many times have been pregnant, RHQ171 - live birth deliveries, RHD280 - had a hysterectomy, RHQ078 - pelvic infection,RHD180 - age at firts livebirth
rhq_data = read_xpt("./data/RHQ_H.xpt") |> select(SEQN,RHQ031,RHQ074,RHQ076, RHQ131,RHQ160, RHQ171, RHD280, RHQ078, RHD180)
# SEx steroide hormone, LBXTST - testosterone (ng/dL), LBXEXT - estradiol(pg/mL), LBXSHGB (nmol/L), promene koncici na LC (0 - above detection limit, 1 below detection limit)
tst_data = read_xpt("./data/TST_H.xpt")
#Phtalates and plasticizesr metabolites
ssphte_data = read_xpt("./data/SSPHTE_H.xpt")

#Personal care and consumer product chemicals and metabolites
ephpp_data = read_xpt("./data/EPHPP_H.xpt")

#phtalates and plasticizers metabolites
phthte_data = read_xpt("./data/PHTHTE_H.xpt")

#alcohol use, ALQ120Q - How often drink alcohol over past 12 months
alq_data = read_xpt("./data/ALQ_H.xpt") |> select(SEQN,ALQ120Q)
#smoking cigarette use, SMQ040 - do you now smoke cigarettes, SMD650 avg cigarettes a day last month
smq_data = read_xpt("./data/SMQ_H.xpt") |> select(SEQN,SMQ040,SMD650)
#medical condition
#mcq_data = read_xpt("./data/MCQ_H.xpt") |> select(SEQN,)
#body measures, BMXBMI - body maxx index
bmx_data = read_xpt("./data/BMX_H.xpt") |> select(SEQN,BMXBMI)
```

Merging datasets

```
#iiner join keep only rows where there is a SEQN match in both datasets
merged_data = inner_join(demo_data, rhq_data, by = "SEQN")
merged_data = inner_join(merged_data, tst_data, by = "SEQN")
merged_data = inner_join(merged_data, ssphte_data, by = "SEQN")
merged_data = inner_join(merged_data, ephpp_data, by = "SEQN")
merged_data = inner_join(merged_data, alq_data, by = "SEQN")
merged_data = inner_join(merged_data, smq_data, by = "SEQN")
merged_data = inner_join(merged_data, bmx_data, by = "SEQN")
merged_data = inner_join(merged_data, phthte_data, by = "SEQN")
nrow(merged_data)
```

```
## [1] 917
```

```r
#jedna promena se vyskytuje ve vice datasetech
merged_data <- merged_data[, !grepl("\\.y$", names(merged_data))]
names(merged_data)[names(merged_data) == "URXUCR.x"] <- "URXUCR"
names(merged_data)[names(merged_data) == "WTSB2YR.x"] <- "WTSB2YR"

var_labels <- sapply(merged_data, function(x) attr(x, "label"))
labels_df <- data.frame(
  variable = names(var_labels),
  label = unname(var_labels)
)
print(labels_df)
```

```
##    variable                                     label
## 1      SEQN              Respondent sequence number
## 2  RIDAGEYR               Age in years at screening
## 3  RIDEXPRG                 Pregnancy status at exam
## 4  DMDEDUC2            Education level - Adults 20+
## 5  RIAGENDR                                   Gender
## 6  INDFMPIR        Ratio of family income to poverty
## 7  WTMEC2YR        Full sample 2 year MEC exam weight
## 8   SDMVPSU               Masked variance pseudo-PSU
## 9  SDMVSTRA           Masked variance pseudo-stratum
## 10   RHQ031    Had regular periods in past 12 months
## 11   RHQ074     Tried for a year to become pregnant?
## 12   RHQ076 Seen a DR b/c unable to become pregnant?
## 13   RHQ131                     Ever been pregnant?
## 14   RHQ160       How many times have been pregnant?
## 15   RHQ171   How many deliveries live birth result?
## 16   RHD280                     Had a hysterectomy?
## 17   RHQ078 Ever treated for a pelvic infection/PID?
## 18   RHD180                   Age at first live birth
## 19   LBXTST              Testosterone, total (ng/dL)
## 20 LBDTSTLC                Testosterone comment code
## 21   LBXEST                      Estradiol (pg/mL)
## 22 LBDESTLC                   Estradiol Comment Code
## 23  LBXSHBG                            SHBG (nmol/L)
## 24 LBDSHGLC                        SHBG Comment Code
## 25 WTSSBH2Y  Surplus specimen B 13-14 2 year weights
## 26   URXUCR                Creatinine, urine (mg/dL)
## 27 SSURMHBP Mono-3-hydroxy-n-butyl phthalate (ng/mL)
## 28 SDUMHBPL     Mono-3-hydroxy-n-butyl phthalate cmt
## 29 SSURHIBP Mono-2-hydroxy-iso-butyl phthlte (ng/mL)
## 30 SDUHIBPL     Mono-2-hydroxy-iso-butyl phthlte cmt
## 31  WTSB2YR                      Subsample B weights
## 32   URXBP3           Urinary Benzophenone-3 (ng/mL)
## 33 URDBP3LC           Urinary Benzophenone-3 comment
## 34   URXBPH              Urinary Bisphenol A (ng/mL)
## 35 URDBPHLC              Urinary Bisphenol A comment
## 36   URXBPF               Urinary Bisphenol F (ug/L)
## 37 URDBPFLC              Urinary Bisphenol F comment
## 38   URXBPS               Urinary Bisphenol S (ug/L)
## 39 URDBPSLC              Urinary Bisphenol S comment
## 40   URXTLC            Urinary Triclocarban (ng/mL)
## 41 URDTLCLC            Urinary Triclocarban comment
## 42   URXTRS               Urinary Triclosan (ng/mL)
## 43 URDTRSLC               Urinary Triclosan comment
## 44   URXBUP                    Butyl paraben (ng/ml)
## 45 URDBUPLC                    Butyl paraben comment
## 46   URXEPB                    Ethyl paraben (ng/ml)
## 47 URDEPBLC                    Ethyl paraben comment
## 48   URXMPB                   Methyl paraben (ng/ml)
## 49 URDMPBLC                   Methyl paraben comment
## 50   URXPPB                   Propyl paraben (ng/ml)
## 51 URDPPBLC                   Propyl paraben comment
## 52   URX14D                2,5-dichlorophenol (ug/L)
## 53 URD14DLC               2,5-dichlorophenol comment
## 54   URXDCB                2,4-dichlorophenol (ug/L)
```

```
## 55 URDDCBLC               2,4-dichlorophenol comment
## 56  ALQ120Q How often drink alcohol over past 12 mos
## 57   SMQ040              Do you now smoke cigarettes
## 58   SMD650 Avg # cigarettes/day during past 30 days
## 59   BMXBMI                  Body Mass Index (kg/m**2)
## 60   URXUCR                  Creatinine, urine (mg/dL)
## 61   URXCNP     Mono(carboxynonyl) Phthalate (ng/mL)
## 62 URDCNPLC    Mono(carboxynonyl) Phthalate cmt code
## 63   URXCOP     Mono(carboxyoctyl) Phthalate (ng/mL)
## 64 URDCOPLC    Mono(carboxyoctyl) phthalate cmt code
## 65   URXECP                     MECP phthalate (ng/mL)
## 66 URDECPLC              MECP phthalate comment code
## 67   URXMBP            Mono-n-butyl phthalate (ng/mL)
## 68 URDMBPLC      Mono-n-butyl phthalate comment code
## 69   URXMC1 Mono-(3-carboxypropyl) phthalate (ng/mL)
## 70 URDMC1LC     Mono-(3-carboxypropyl) phthalate cmt
## 71   URXMEP              Mono-ethyl phthalate (ng/mL)
## 72 URDMEPLC        Mono-ethyl phthalate comment code
## 73   URXMHH                     MEHP phthalate (ng/mL)
## 74 URDMHHLC              MEHP phthalate comment code
## 75  URXMHNC                              MHNCH (ng/mL)
## 76 URDMCHLC                       MHNCH comment code
## 77   URXMHP   Mono-(2-ethyl)-hexyl phthalate (ng/mL)
## 78 URDMHPLC  Mono-(2-ethyl)-hexyl phthalate cmt code
## 79   URXMIB          Mono-isobutyl phthalate (ng/mL)
## 80 URDMIBLC     Mono-isobutyl phthalate comment code
## 81   URXMNP          Mono-isononyl phthalate (ng/mL)
## 82 URDMNPLC     Mono-isononyl phthalate comment code
## 83   URXMOH                     MEOH phthalate (ng/mL)
## 84 URDMOHLC              MEOH phthalate comment code
## 85   URXMZP            Mono-benzyl phthalate (ng/mL)
## 86 URDMZPLC       Mono-benzyl phthalate comment code
```

Remove, based on exclusion, inclusion criteria from the plan

```
#age between 18 and 45 and female only, not pregnant
clean_data <- merged_data[merged_data$RIDAGEYR >= 18 & merged_data$RIDAGEYR <= 50 & merged_da
ta$RIAGENDR == 2 & merged_data$RIDEXPRG == 2,]
nrow(clean_data)
```

```
## [1] 460
```

```
#extreme urinary creatine values

clean_data <- clean_data[clean_data$URXUCR<300,]

#missing key covariates -> R si NaN values odstraní automaticky pri vytvareni modelu ?

#urinary chemicals measurement and MEC examination
clean_data <- clean_data[clean_data$WTSSBH2Y>0 & clean_data$WTSB2YR>0 & clean_data$WTMEC2YR>
0,]
nrow(clean_data)
```

```
## [1] 448
```

```
#at least one detectable urinary chemical measurement (ze vsech chemikalii v EPHPP a SSPHTE)
all_vars <- union(union(names(ssphte_data), names(ephpp_data)),names(phthte_data))
exclude_vars <- c( "SEQN","WTSSBH2Y","WTSB2YR" )
edc_vars <- setdiff(all_vars, exclude_vars)
length(edc_vars)
```

```
## [1] 55
```

```
data_1 <- clean_data[
  rowSums(!is.na(clean_data[, edc_vars])) >= 1, #alespon jedna hodnota ze vsech merenych chem
ikalii neni Nan (missing)
]
# at least one detectable hormone measurement (not missing - nan)
horm_vars <- c("LBXSHBG","LBXEST","LBXTST")
data_1 <- data_1[
  rowSums(!is.na(data_1[, horm_vars])) >= 1, #alespon jedna hodnota ze vsech hormonu neni nan
]

nrow(data_1)
```

```
## [1] 324
```

Survey design object -> to work with the weights

NHANES: You must use the weight of the smallest subpopulation that includes all the variables you want to include in your analysis. Z tech mensich souboru obsahují speciální vahy jenom ty chemikálie (EPHPP a SSPHTE a PHTHTE). EPHPP a PHTHTE maji stejne (WTSB2YR) a SSPHTE ma jiné (WTSSBH2Y). Nevim ktera z nich by se mela pouzit, ale nejmensi je dataset SSPHTE.

```
# v ukazkach pak na vsechny upravy pouzivaji uz jenom survey desing objekt -> vytvorit z cely
ch dat a znovu stejne profiltrovat
data_w <- svydesign(id = ~SDMVPSU, strata= ~SDMVSTRA, weight=~WTSB2YR,   nest=TRUE, data=merg
ed_data)

obj_cleaned <- data_w |>
  subset(
    RIDAGEYR >= 18 &
    RIDAGEYR <= 50 &
    RIAGENDR == 2 &
    RIDEXPRG == 2 &
    URXUCR < 300 &
    WTSSBH2Y > 0 &
    WTSB2YR > 0 &
    WTMEC2YR > 0 &
    rowSums(!is.na(as.data.frame(data_w$variables[, edc_vars]))) >= 1 &
    rowSums(!is.na(as.data.frame(data_w$variables[, horm_vars]))) >= 1
  )
nrow(obj_cleaned$variables)
```

```
## [1] 324
```

```
summary(obj_cleaned)
```

```
## Stratified 1 - level Cluster Sampling design (with replacement)
## With (30) clusters.
## subset(data_w, RIDAGEYR >= 18 & RIDAGEYR <= 50 & RIAGENDR ==
##     2 & RIDEXPRG == 2 & URXUCR < 300 & WTSSBH2Y > 0 & WTSB2YR >
##     0 & WTMEC2YR > 0 & rowSums(!is.na(as.data.frame(data_w$variables[,
##     edc_vars]))) >= 1 & rowSums(!is.na(as.data.frame(data_w$variables[,
##     horm_vars]))) >= 1)
## Probabilities:
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
## 2.515e-06 6.348e-06 1.011e-05 1.047e-05 1.392e-05 3.033e-05
## Stratum Sizes:
##            104 105 106 107 108 109 110 111 112 113 114 115 116 117 118
## obs         13  18  20  26  20  21  15  26  29  22  27  23  20  17  27
## design.PSU   2   2   2   2   2   2   2   2   2   2   2   2   2   2   2
## actual.PSU   2   2   2   2   2   2   2   2   2   2   2   2   2   2   2
## Data variables:
##  [1] "SEQN"     "RIDAGEYR" "RIDEXPRG" "DMDEDUC2" "RIAGENDR" "INDFMPIR"
##  [7] "WTMEC2YR" "SDMVPSU"  "SDMVSTRA" "RHQ031"   "RHQ074"   "RHQ076"
## [13] "RHQ131"   "RHQ160"   "RHQ171"   "RHD280"   "RHQ078"   "RHD180"
## [19] "LBXTST"   "LBDTSTLC" "LBXEST"   "LBDESTLC" "LBXSHBG"  "LBDSHGLC"
## [25] "WTSSBH2Y" "URXUCR"   "SSURMHBP" "SDUMHBPL" "SSURHIBP" "SDUHIBPL"
## [31] "WTSB2YR"  "URXBP3"   "URDBP3LC" "URXBPH"   "URDBPHLC" "URXBPF"
## [37] "URDBPFLC" "URXBPS"   "URDBPSLC" "URXTLC"   "URDTLCLC" "URXTRS"
## [43] "URDTRSLC" "URXBUP"   "URDBUPLC" "URXEPB"   "URDEPBLC" "URXMPB"
## [49] "URDMPBLC" "URXPPB"   "URDPPBLC" "URX14D"   "URD14DLC" "URXDCB"
## [55] "URDDCBLC" "ALQ120Q"  "SMQ040"   "SMD650"   "BMXBMI"   "URXUCR"
## [61] "URXCNP"   "URDCNPLC" "URXCOP"   "URDCOPLC" "URXECP"   "URDECPLC"
## [67] "URXMBP"   "URDMBPLC" "URXMC1"   "URDMC1LC" "URXMEP"   "URDMEPLC"
## [73] "URXMHH"   "URDMHHLC" "URXMHNC"  "URDMCHLC" "URXMHP"   "URDMHPLC"
## [79] "URXMIB"   "URDMIBLC" "URXMNP"   "URDMNPLC" "URXMOH"   "URDMOHLC"
## [85] "URXMZP"   "URDMZPLC"
```

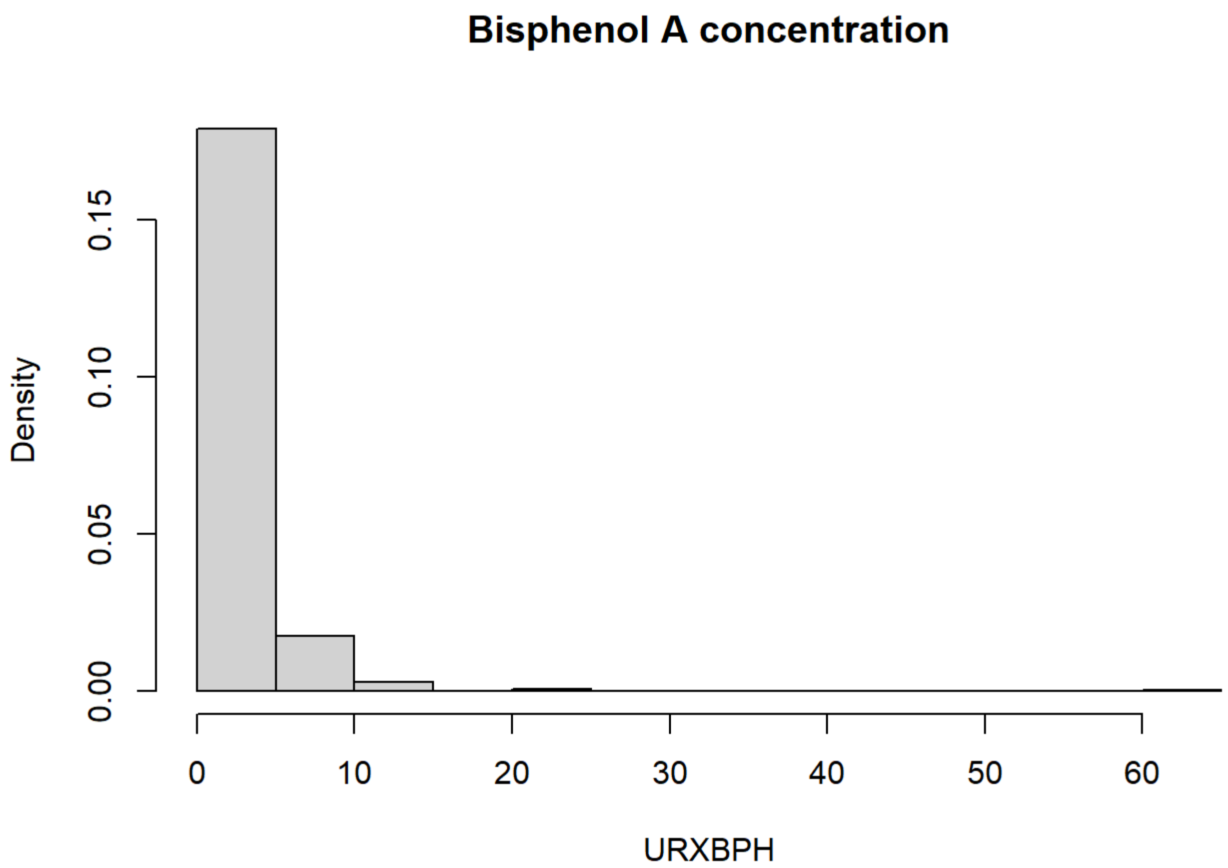
```
#vytvoreny survey objekt porad obsahuje sloupce s vahama z vetsich datasetu -> pri vytvareni
modelu upravit a nebrat je v uvahu
```

Variables statistics

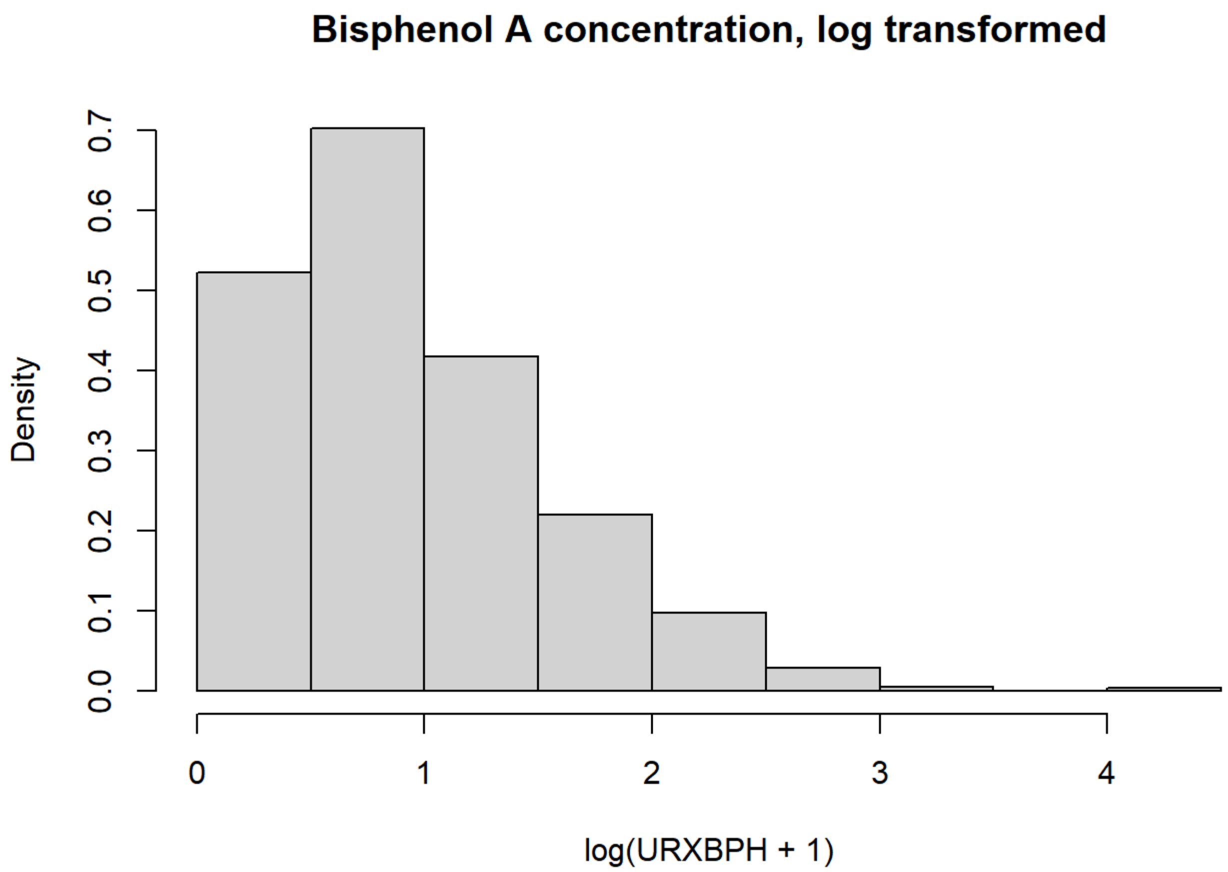
```
#age
svymean(~RIDAGEYR,obj_cleaned,na.rm = TRUE)
```

```
##            mean     SE
## RIDAGEYR 32.326 0.7096
```

```
 #visualize
svyhist(~URXBPH, design=obj_cleaned, main="Bisphenol A concentration")
```




```
svyhist(~log(URXBPH + 1), design=obj_cleaned, main="Bisphenol A concentration, log transforme
d")
```



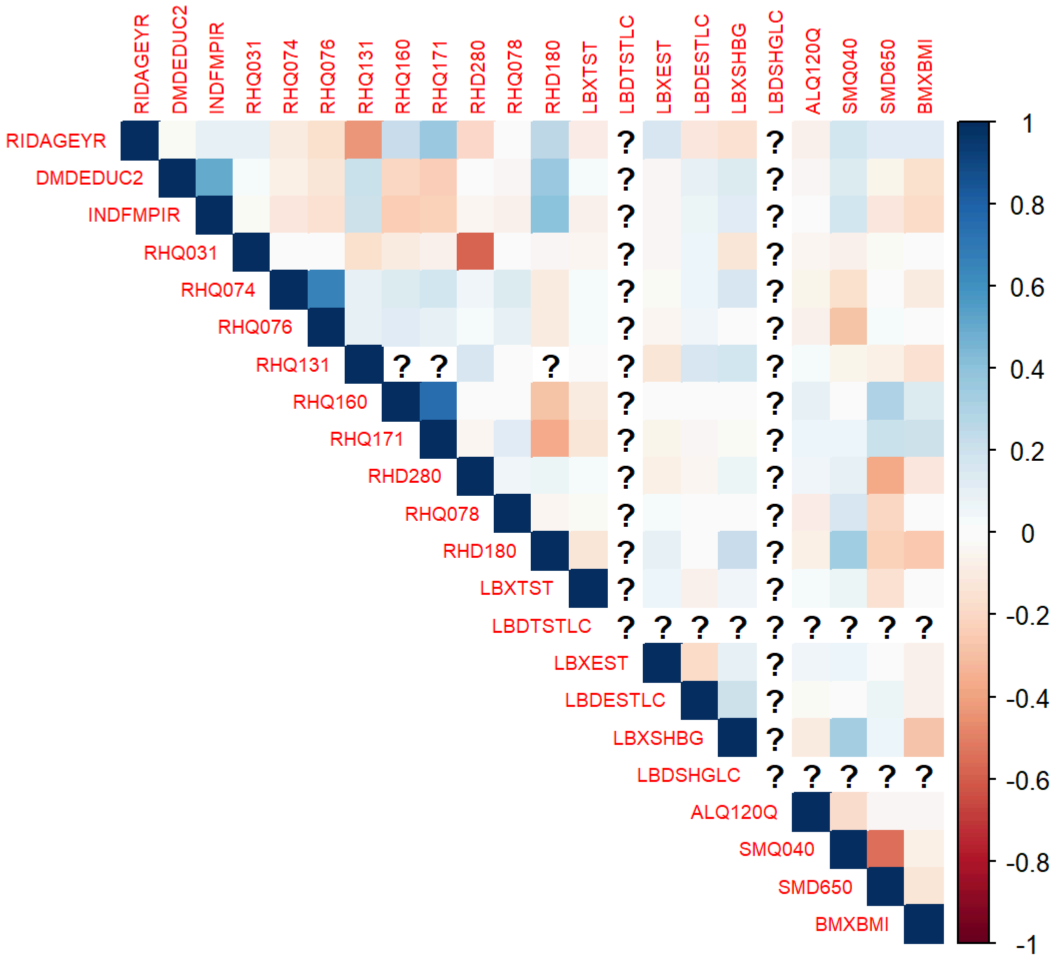
```
#Correlation matrix - na vsem krome chemikalii, weights a gender a pregnant, protoze to jsou po filtraci vlastne konstantni sloupce pro vsechny stejne
exclude_vars <- c("SEQN","WTMEC2YR", "WTSSBH2Y", "WTSB2YR",edc_vars,"SDMVPSU","SDMVSTRA","RIDEXPRG","RIAGENDR")
cor_vars <- setdiff(names(obj_cleaned$variables), exclude_vars)
length(cor_vars)
```

```
## [1] 22
```

```
cor_matrix <- cor(obj_cleaned$variables[, cor_vars],use = "pairwise.complete.obs")
```

```
## Warning in cor(obj_cleaned$variables[, cor_vars], use =
## "pairwise.complete.obs"): the standard deviation is zero
```

```
corrplot::corrplot(cor_matrix,tl.cex = 0.6,method = "color",type = "upper")
```



```
#V clanku: Use PCA to deal with collinearity of chemicals -> get fewer independent component
s. Factors with eigen values higher than 1 were determined as the main components
#pridat dalsi sloupce do datasetu s logtransformovanymi chemikaliemi, edc_vars obsahuje jmena
sloupcu vsech chemikalii v datasetu
obj_cleaned$variables[, paste0("log_", edc_vars)] <-log(obj_cleaned$variables[, edc_vars] +
1)
log_edc_vars <- paste0("log_", edc_vars) #log_edc_vars -> jmena vsech log transformovanych ch
emikalii


#potreba odstranit tuto chemikalii, jinak error: cannot rescale constant/zero column to unit
variance, je to comment code (value 0 - at or above detection limit)
del <- c( "log_URDMEPLC" )
log_edc_vars <- setdiff(log_edc_vars,del)


pca_w <- svyprcomp(
  ~ as.matrix(obj_cleaned$variables[, log_edc_vars]),
  design = obj_cleaned,
  center = TRUE,
  scale  = TRUE
)

summary(pca_w)
```

```
## Importance of components:
##                           PC1     PC2     PC3     PC4     PC5    PC6    PC7
## Standard deviation     3.6704 2.21698 1.66427 1.60227 1.46470 1.3630 1.2835
## Proportion of Variance 0.2495 0.09102 0.05129 0.04754 0.03973 0.0344 0.0305
## Cumulative Proportion  0.2495 0.34050 0.39179 0.43933 0.47906 0.5135 0.5440
##                            PC8     PC9    PC10    PC11    PC12    PC13   PC14
## Standard deviation     1.22780 1.17369 1.16733 1.11589 1.09123 1.07147 1.0236
## Proportion of Variance 0.02792 0.02551 0.02523 0.02306 0.02205 0.02126 0.0194
## Cumulative Proportion  0.57189 0.59740 0.62263 0.64569 0.66774 0.68900 0.7084
##                           PC15    PC16    PC17    PC18    PC19    PC20    PC21
## Standard deviation     1.00989 0.97708 0.97394 0.95557 0.94227 0.88011 0.85459
## Proportion of Variance 0.01889 0.01768 0.01757 0.01691 0.01644 0.01434 0.01352
## Cumulative Proportion  0.72729 0.74497 0.76254 0.77945 0.79589 0.81023 0.82376
##                           PC22    PC23    PC24    PC25    PC26   PC27    PC28
## Standard deviation     0.79780 0.77014 0.74809 0.73141 0.72127 0.7048 0.68712
## Proportion of Variance 0.01179 0.01098 0.01036 0.00991 0.00963 0.0092 0.00874
## Cumulative Proportion  0.83555 0.84653 0.85689 0.86680 0.87643 0.8856 0.89438
##                           PC29    PC30    PC31    PC32    PC33    PC34   PC35
## Standard deviation     0.67646 0.66506 0.65618 0.60972 0.59421 0.58080 0.5691
## Proportion of Variance 0.00847 0.00819 0.00797 0.00688 0.00654 0.00625 0.0060
## Cumulative Proportion  0.90285 0.91104 0.91902 0.92590 0.93244 0.93868 0.9447
##                           PC36    PC37    PC38    PC39    PC40   PC41    PC42
## Standard deviation     0.55169 0.53131 0.52073 0.51261 0.50078 0.4761 0.46783
## Proportion of Variance 0.00564 0.00523 0.00502 0.00487 0.00464 0.0042 0.00405
## Cumulative Proportion  0.95032 0.95555 0.96057 0.96543 0.97008 0.9743 0.97833
##                           PC43    PC44    PC45    PC46    PC47    PC48    PC49
## Standard deviation     0.45673 0.41925 0.40815 0.36105 0.32378 0.30685 0.27925
## Proportion of Variance 0.00386 0.00326 0.00309 0.00241 0.00194 0.00174 0.00144
## Cumulative Proportion  0.98219 0.98545 0.98853 0.99095 0.99289 0.99463 0.99608
##                           PC50    PC51   PC52    PC53    PC54
## Standard deviation     0.27372 0.24625 0.1942 0.15436 0.12161
## Proportion of Variance 0.00139 0.00112 0.0007 0.00044 0.00027
## Cumulative Proportion  0.99746 0.99859 0.9993 0.99973 1.00000
```

```
eigenvalues <- pca_w$sdev^2
prop_var    <- eigenvalues / sum(eigenvalues)
cum_var     <- cumsum(prop_var)

pca_table <- data.frame(
  PC = paste0("PC", seq_along(eigenvalues)),
  Eigenvalue = eigenvalues,
    Proportion_of_variance = prop_var,
  Cumulative_variance = cum_var
)

pca_table
```

```
##      PC  Eigenvalue Proportion_of_variance Cumulative_variance
## 1   PC1 13.47199921           0.2494814668           0.2494815
## 2   PC2  4.91501617           0.0910188180           0.3405003
## 3   PC3  2.76978822           0.0512923744           0.3917927
## 4   PC4  2.56726255           0.0475418991           0.4393346
## 5   PC5  2.14535004           0.0397287045           0.4790633
## 6   PC6  1.85783366           0.0344043270           0.5134676
## 7   PC7  1.64725265           0.0305046786           0.5439723
## 8   PC8  1.50748634           0.0279164138           0.5718887
## 9   PC9  1.37753841           0.0255099706           0.5973987
## 10 PC10  1.36265086           0.0252342752           0.6226329
## 11 PC11  1.24522132           0.0230596541           0.6456926
## 12 PC12  1.19078172           0.0220515133           0.6677441
## 13 PC13  1.14805364           0.0212602525           0.6890043
## 14 PC14  1.04770672           0.0194019763           0.7084063
## 15 PC15  1.01988194           0.0188867026           0.7272930
## 16 PC16  0.95469474           0.0176795322           0.7449726
## 17 PC17  0.94856815           0.0175660768           0.7625386
## 18 PC18  0.91311908           0.0169096126           0.7794482
## 19 PC19  0.88787004           0.0164420378           0.7958903
## 20 PC20  0.77458899           0.0143442406           0.8102345
## 21 PC21  0.73032180           0.0135244777           0.8237590
## 22 PC22  0.63648907           0.0117868346           0.8355458
## 23 PC23  0.59312308           0.0109837607           0.8465296
## 24 PC24  0.55964367           0.0103637717           0.8568934
## 25 PC25  0.53496044           0.0099066749           0.8668000
## 26 PC26  0.52022396           0.0096337771           0.8764338
## 27 PC27  0.49677572           0.0091995504           0.8856334
## 28 PC28  0.47213324           0.0087432082           0.8943766
## 29 PC29  0.45760456           0.0084741585           0.9028507
## 30 PC30  0.44230836           0.0081908955           0.9110416
## 31 PC31  0.43057273           0.0079735690           0.9190152
## 32 PC32  0.37175610           0.0068843722           0.9258996
## 33 PC33  0.35308557           0.0065386217           0.9324382
## 34 PC34  0.33732664           0.0062467896           0.9386850
## 35 PC35  0.32390553           0.0059982505           0.9446832
## 36 PC36  0.30436391           0.0056363687           0.9503196
## 37 PC37  0.28228542           0.0052275077           0.9555471
## 38 PC38  0.27116462           0.0050215671           0.9605687
## 39 PC39  0.26276932           0.0048660985           0.9654348
## 40 PC40  0.25077733           0.0046440247           0.9700788
## 41 PC41  0.22667565           0.0041976973           0.9742765
## 42 PC42  0.21886869           0.0040531238           0.9783296
## 43 PC43  0.20860520           0.0038630593           0.9821927
## 44 PC44  0.17577010           0.0032550018           0.9854477
## 45 PC45  0.16659021           0.0030850039           0.9885327
## 46 PC46  0.13035903           0.0024140562           0.9909467
## 47 PC47  0.10483109           0.0019413166           0.9928881
## 48 PC48  0.09415453           0.0017436024           0.9946317
## 49 PC49  0.07797950           0.0014440649           0.9960757
## 50 PC50  0.07492502           0.0013875003           0.9974632
## 51 PC51  0.06063661           0.0011229001           0.9985861
## 52 PC52  0.03773256           0.0006987512           0.9992849
## 53 PC53  0.02382795           0.0004412584           0.9997261
## 54 PC54  0.01478834           0.0002738581           1.0000000
```